



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHANMUGA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

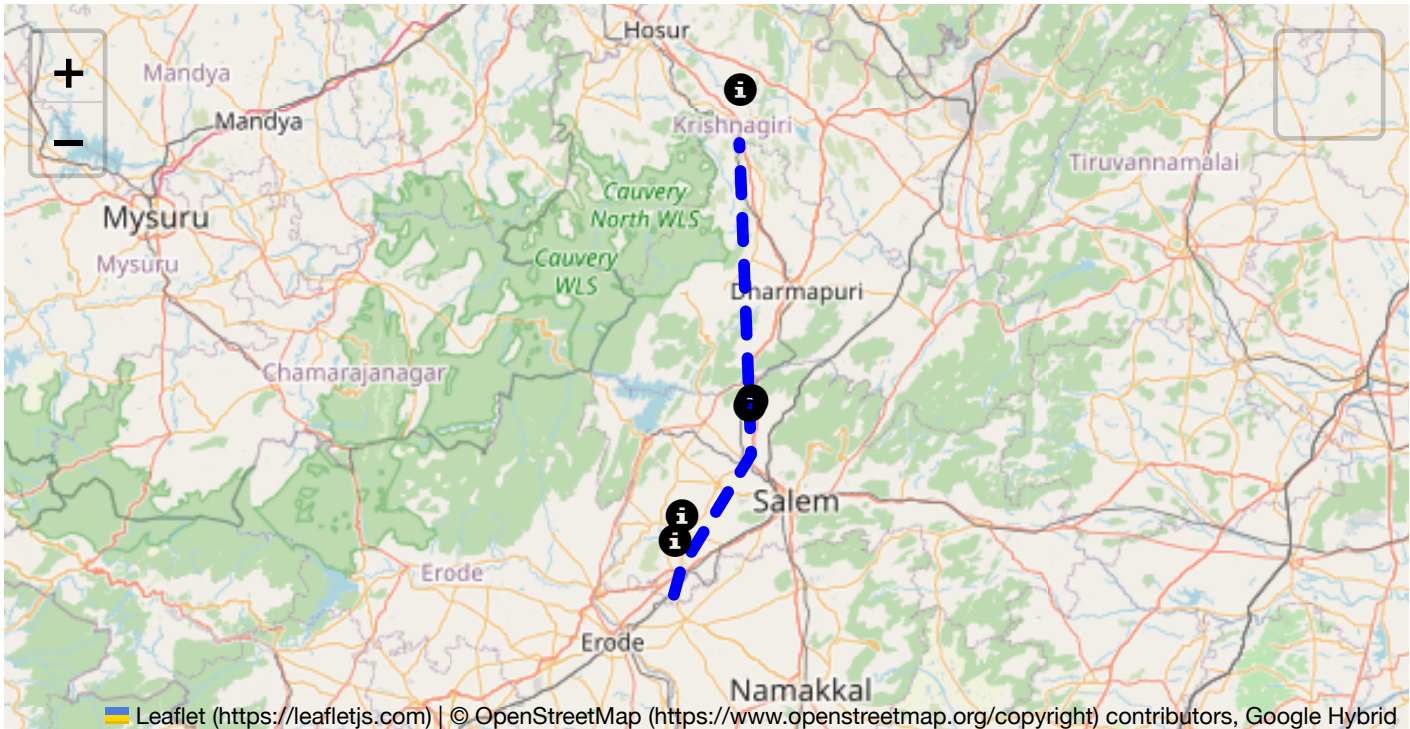
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

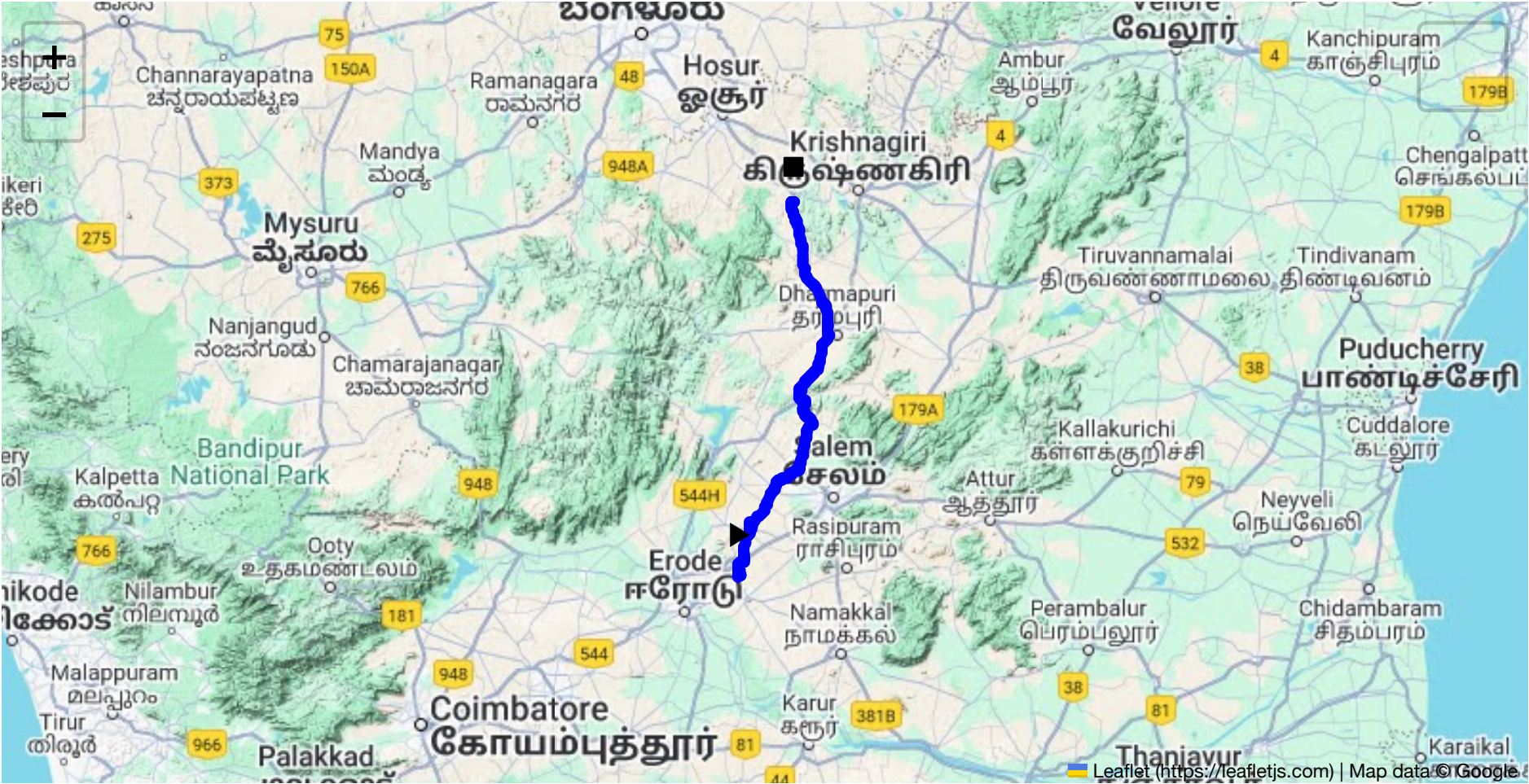
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 142.45 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.6 hours

Start: (11.4381, 77.8734)
End: (12.476537, 78.026588)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W in Sangagiri to F2GG+XPV in Rayakottai spans about 142.45 kilometers, passing through Salem, Kottaimettupatti, and Omalur. It primarily utilizes major highways, including local roads, before reaching the final destination in Rayakottai. This route is characterized by both urbanized areas and rural stretches.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with hot summers, substantial monsoon rains between June and September, and mild winters. Along this route, drivers should anticipate heavy rainfall during monsoon seasons, which could lead to flooding or slick road surfaces, particularly in low-lying areas.

3. Traffic Patterns and Congestion-Prone Areas

- Salem:** Being a major city, Salem experiences significant traffic, especially during morning (8-10 AM) and evening (5-7 PM) peak hours. Congestion is more prevalent in commercial zones.
- Omalur and Kottaimettupatti:** These areas generally have moderate traffic, but congestion may occur during school hours and local market days.

4. Road Quality and Infrastructure

- Highways:** Generally well-maintained with good surface quality, suitable for heavy vehicles.
- Rural sections:** May have narrower roads with less reliable maintenance, posing potential risks for larger trucks.

5. Suggestions for Alternative Routes for Emergencies

In case of road closures or severe congestion, consider:

- Bypassing Salem city by taking peripheral roads.
- Utilizing NH 44 for bypassing minor towns if conditions in Kottaimettupatti or Omalur become problematic.

6. Local Regulations Affecting Hazardous Material Transport

Tamil Nadu mandates that transport of hazardous materials adhere to stringent guidelines, including travel during prescribed hours, adequate labeling, and adherence to designated routes. Violations can result in hefty fines and increased inspection.

7. Overview of Historical Incidents Involving Heavy Vehicles

There have been instances of accidents on narrow and poorly lit stretches, primarily at night. Salem city has also reported incidents involving trucks due to high congestion, frequently attributed to poor visibility and driver fatigue.

8. Environmental Considerations and Sensitive Areas

- **Forested Areas:** Portions of the route may run adjacent to reserved forest areas; care must be taken to minimize noise and avoid pollution.
- **Water Bodies:** Crossing near water sources necessitates precautions to avoid contamination in the event of a spillage.

9. Communication Coverage

While most of the route enjoys decent cellular coverage, rural areas between Omalur and Rayakottai may experience intermittent signal issues. Drivers should ensure offline maps and pre-downloaded routes are available as backups.

10. Estimated Emergency Response Times

- **Urban Areas (Salem):** 15-30 minutes due to proximity to major facilities.
- **Rural Sections:** 30-60 minutes, with potential delays exacerbated by poor accessibility or weather conditions.

11. Environmental Considerations and Sensitive Areas

As mentioned earlier, close proximity to forested areas and water bodies along parts of the route necessitates careful navigation to prevent environmental harm. Adherence to environmental regulations regarding emissions and spillage is critical.

12. Overall Summary of Risk Assessment

The route presents moderate risks associated with traffic congestion, weather conditions, and road quality. Proper precautions, adherence to regulations, and thorough pre-trip planning can mitigate potential issues. It's essential for drivers to remain alert, particularly during adverse weather conditions, and maintain communication readiness in areas with known signal issues. Emergency plans should

include predetermined alternative routes and knowledge of local emergency contacts to ensure swift response in case of incidents.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.7520621, 78.0482089	10 KM/Hr
1	U-Turn	High	11.7460352, 78.0474485	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.48363, 77.88780	30 KM/Hr
9	Turn	Medium	11.49182, 77.88762	30 KM/Hr
10	Turn	High	11.49451, 77.88489	15 KM/Hr
11	Turn	Medium	11.55573, 77.89751	30 KM/Hr
12	Turn	Medium	11.55640, 77.89913	30 KM/Hr
13	Turn	Medium	11.56627, 77.90652	30 KM/Hr
14	Turn	Medium	11.57677, 77.90863	30 KM/Hr
15	Turn	Medium	11.58249, 77.90829	30 KM/Hr
16	Turn	Medium	11.58294, 77.90775	30 KM/Hr
17	Turn	Medium	11.58315, 77.90685	30 KM/Hr
18	Turn	Medium	11.58427, 77.90602	30 KM/Hr
19	Turn	Medium	11.59297, 77.90746	30 KM/Hr
20	Turn	Medium	11.63836, 77.94819	30 KM/Hr
21	Turn	High	11.64176, 77.95188	15 KM/Hr
22	Turn	Medium	11.64660, 77.95114	30 KM/Hr
23	Turn	Medium	11.67870, 77.96848	30 KM/Hr
24	Turn	Medium	11.70120, 77.97940	30 KM/Hr
25	Turn	High	11.70233, 77.97869	15 KM/Hr
26	Turn	Medium	11.71721, 77.99982	30 KM/Hr
27	Turn	Medium	11.71923, 78.01140	30 KM/Hr
28	Turn	Medium	11.73778, 78.03472	30 KM/Hr
29	Turn	Medium	11.73654, 78.03858	30 KM/Hr
30	Turn	Medium	11.73665, 78.03907	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	11.74226, 78.04688	30 KM/Hr
32	Blind Spot	Blind Spot	11.75207, 78.04830	10 KM/Hr
33	Blind Spot	Blind Spot	11.74626, 78.04746	10 KM/Hr
34	Turn	Medium	11.74633, 78.04740	30 KM/Hr
35	Turn	Medium	11.86873, 78.08374	30 KM/Hr
36	Turn	Medium	11.87966, 78.08187	30 KM/Hr
37	Turn	Medium	11.92609, 78.06513	30 KM/Hr
38	Turn	Medium	11.93719, 78.05372	30 KM/Hr
39	Turn	High	11.95828, 78.04983	15 KM/Hr
40	Turn	Medium	11.95997, 78.04946	30 KM/Hr
41	Turn	Medium	11.96335, 78.05141	30 KM/Hr
42	Turn	High	11.96408, 78.05131	15 KM/Hr
43	Turn	Medium	11.96511, 78.04966	30 KM/Hr
44	Turn	High	11.96608, 78.04919	15 KM/Hr
45	Turn	Medium	11.96819, 78.05088	30 KM/Hr
46	Turn	Medium	12.12010, 78.13282	30 KM/Hr
47	Turn	High	12.49630, 78.02711	15 KM/Hr
48	Blind Spot	Blind Spot	12.49616, 78.02929	10 KM/Hr
49	Turn	Medium	12.49690, 78.02915	30 KM/Hr
50	Blind Spot	Blind Spot	12.49791, 78.02771	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
9	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
10	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
12	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
13	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
14	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
15	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
16	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
17	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
21	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
22	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
25	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
26	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
27	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
28	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
29	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
30	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
32	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
33	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
34	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
19	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
31	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.7520621, 78.0482089



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 11.7460352, 78.0474485



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48363, 77.88780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49182, 77.88762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 77.88489



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55573, 77.89751



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58315, 77.90685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58427, 77.90602



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59297, 77.90746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63836, 77.94819



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64176, 77.95188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64660, 77.95114



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67870, 77.96848



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70120, 77.97940



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70233, 77.97869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71721, 77.99982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71923, 78.01140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73778, 78.03472



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73654, 78.03858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73665, 78.03907



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74226, 78.04688



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75207, 78.04830



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.74626, 78.04746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74633, 78.04740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12010, 78.13282



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.49630, 78.02711



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.49616, 78.02929



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.49690, 78.02915



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.49791, 78.02771
